V3.2 | Common Boat-Critical Components

Rationale-Led ArduPilot Rover Platform: Verified Specifications and Integration Gates

Year 3 Engineering Group Project | Microplastic Mapping

5 November 2026 | Engineering rationale, sourcing evidence and test gates

Scope rule. This document contains the **boat common core only**. Track A (live flow cell) and Track B (sample collection) have independent sensors, compute, pumping and payload budgets. The boat flight controller and the payload microcontroller exchange time, position, status and commands across a defined link. No payload hardware is counted in the boat cost.

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1 Requirements, design basis and procurement discipline

The supplied V3 Method Rationale defines a 20 m by 20 m area, a 120 min total mission and Gate G9 for adequate flotation and energy [1]. Those are **mission targets from the supplied project source**, not observed capabilities. The original ArduPilot boat tutorial is a *conceptual reference* rather than a component or weight specification [2]. The goal is a lighter differential-thrust twin-pontoon Rover, consistent with ArduPilot's boat and motor-configuration guides [3], [4].

Purchasing rule: every proposed purchasable item links to a registered supplier (Cytron or GI Electronic) or Shopee Malaysia; no manufacturer or overseas website is used as the purchase channel. Manufacturer/ArduPilot pages may be cited solely as *technical evidence*. Prices are RM displayed on 5 November 2026 or identified V2.4 source allowances. Seller variant prices, stock and cable connectors may change. **NR = not reported/verified for the sold SKU**; **A = allowance** (not a verified quote). The subtotal *cannot* be interpreted as a finished approved procurement quote. Evidence classes used below: **M** = manufacturer / official ArduPilot technical documentation; **S** = registered-supplier specification; **L** = Shopee listing; **D** = calculated design inference. These classes should not be treated as equally authoritative.

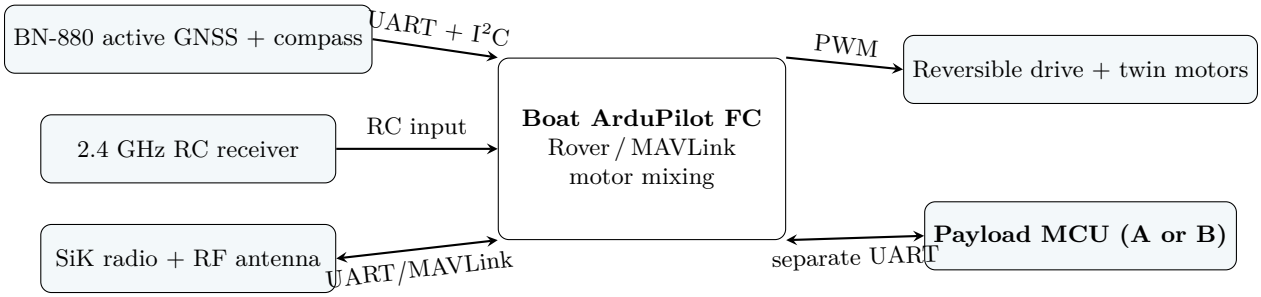
The overall budget constraint is **RM2,000**; the design should retain headroom instead of spending the entire ceiling. No 1–16 kg useful payload category is assigned to these configurations: physical payload mass and sensor power remain Track-dependent and must be separately measured. The COG, flotation, peak current, runtime and GNSS precision have not passed any physical test.

2 Antenna roles and communication architecture

Link	Purpose and component	What it does not do
GNSS receive	**Active GNSS patch/LNA** in Beitian BN-880 plus compass; receives satellite signals for positioning and heading calculations [5], [6], [7].	*Receive only:* cannot receive boat commands or send boat telemetry.
Manual RC	Dedicated **2.4 GHz** handheld transmitter + receiver, normally used for human take-over, mode switching and safety stop. HotRC bundle lists six channels and receiver included [8], [9]. **Compatibility NOT yet proven:** a receiver offering individual PWM channel outputs needs a verified PPM encoder or a different receiver supplying SBUS/PPM/serial RC [10], [11].	RC channels are *not* automatically a Mission Planner MAVLink ground-station link.
Bidirectional RF	**SiK-style UART telemetry pair** with a boat antenna and ground antenna: carries MAVLink status and commands between Rover and a computer [12], [13], [14].	Not a substitute for the independent manual RC link until a tested fail-safe architecture has been approved.
Onboard Wi-Fi	CoreWing Mini V2 includes onboard Wi-Fi access-point/STA capability and supports Mission Planner TCP or UDP telemetry without a separate SiK transceiver [10].	A cost-saving *short-range development option*, not a verified outdoor communication link. No operating radius, waterproof antenna or drop-out performance is substantiated.

Cost-saving link variant. CoreWing Wi-Fi is already integrated in the budgeted flight controller [10]; omitting the unverified **RM160** SiK-pair allowance from Tier A reduces its *conditional* electronics-and-hull subtotal from RM1,192.37 to **RM1,032.37**, leaving **RM967.63** against the RM2,000 ceiling. This is an arithmetic planning comparison, *not* a tested, complete boat, and the independent 2.4 GHz RC link remains mandatory. The tier totals later retain SiK for comparison.

Active GPS detail. BN-880 is marketed with a patch antenna and compass; a BN-880 series technical sheet identifies an *active* antenna, but variant/genuine-module verification is required before purchase [5], [6], [7]. If a truly separate weatherproof GPS antenna is needed, GI Electronic's active antenna is available for **RM27**, claims 26 dB gain, 50 Ω impedance and 18 mA typical draw, and includes an adapter cable [15]. **Do not add it directly to BN-880:** an appropriate RF connector, GNSS receiver with active antenna bias voltage, water seal and cable-loss assessment are prerequisites; GI's antenna is an *alternative RF front end*, not an automatic performance upgrade.



UART allocation gate: dedicate one logical serial port to GNSS, one to SiK telemetry and one to payload MCU MAVLink, while RC uses its documented receiver interface. CoreWing's GPS, telemetry, external-compass and RC connectors are documented; **exact free UART pins, connector voltage and firmware build must be checked on the purchased board** [10]. MAVLink messages may be dropped, so synchronise timestamps and handle missing observations explicitly [16]. Use separate antenna positions and isolate high-current motor wiring from compass and RF leads.

3 Flight-controller comparison and SpeedyBee decision

ECU / purchase	Price, mass and power	Navigation / control evidence	Decision
CoreWing F405 Wing Mini V2 [17]	**RM264.43** voucher listing; 19 g board-family published figure; 2–6S [10], [17]	ArduPilot Rover board documentation lists GPS, compass, telemetry, receiver, motor outputs and integrated BEC; connector voltage and servo-rail BEC warnings apply [10].	**Preferred in tiers A–C.** UART/pin check required.
SpeedyBee F405 V4 **60A quad ESC stack** [18]	**RM445.40**; FC: 41.6 × 39.4 × 7.8 mm (10.5 g). ESC: 45.6 × 44 × 8 mm (23.5 g) [19]	FC has ArduPilot Rover support, 9 PWM outputs, 6 serial interfaces (some reserved), external I^2C compass and microSD [19], [20].	**Comparator only.** ESC is quadcopter BLHeli_S; reversible boat operation *not confirmed.*

Why not buy the suggested stack first? The SpeedyBee ESC's 60 A channel rating is a *current limit* rather than a motor-power requirement, and its four-channel fixed-wing/quadcopter packaging creates no advantage for a two-motor boat. More importantly, ArduPilot's *bidirectional DShot telemetry* is **not the same** as physical forward/reverse propulsion; reversal depends on the ESC firmware/mode and motor protocol, and cannot be assumed from the published stack specification [4], [19], [20]. The stack may be retested later but is not costed as a proven reversing drive.

4 Propulsion and motor-driver technical comparison

Table 1: Motor and driver options: every rating is source-linked; unreported parameters are not inferred. Electrical input is NOT shaft output or horsepower.

Model/channel	**Supply**	**Dimensions**	**Mass**	**Thrust / shaft W / hp**	**Under-load power/current**	**Max current**	**Price each**	**Purchase; technical evidence**
12V three-blade jet unit, Shopee [21]	12V advertised	NR, sold variant	NR	thrust/shaft W/hp **NR**	current under water **NR**	stall/max **NR**	RM38.50–74.80 variant range; planning **74.80**	Shopee; no reliable maker sheet; motor type *unconfirmed*
Hobbyocean T60 + 40A ESC variations [22], [23]	12–16V and 2–4S *claimed by third-party spec*	NR for purchased SKU	NR	seller/third-party 300W electrical-rated class; thrust/hp for SKU **NR**	third-party T60 15–23A *claimed*; no certified curve	ESC 40A *variant claim*; motor limit NR	RM225.98–321.46 variants; **321.46 A**	Shopee; third-party tech, not procurement; verify *full kit*
M060 waterproof BLDC thruster [24], [25]	12–24V advertised	**NR exact SKU**	**NR exact SKU**	seller variant thrust/shaft W/hp **NR**; maker-family up-to-3kgf *not transferable*	max-loaded W/A exact SKU **NR**	max continuous/stall **NR**	**273.32**, voucher listing	Shopee; manufacturer-family reference
Cytron MDD10A, *brushed only* [26]	5–30V motor	84.5× 62 mm [26]	NR	n/a (H-bridge driver)	loss/heat case-specific	**10A continuous / channel**, check product manual; 2 channels	**108.78**	Cytron; NOT direct RC pulse input
Cytron MDD3A, *brushed only* [27]	4–16V motor	NR	NR	n/a	current load user-dependent	**3A continuous / channel, 5A peak**; 2 channels	**39.06**	Cytron; *not* safely sized without motor current
ZTW Shark G2 20A reversible ESC [28], [29]	2–4S advertised for 20A family	maker data for one variant; sold SKU NR	NR sold SKU	n/a; supports BLDC only	loss/thermal draw **NR**	20A *nominal selected variant*, actual motor matching unproven	**100.00 A**, seller listing RM67.65–227.70	Shopee; OEM family page

Motor-drive incompatibility gates. Tier A may use MDD10A *only after* a two-wire brushed motor is positively identified, its submerged load is measured below the channel limit and the controller input is translated into PWM/direction. BLDC motors require BLDC ESCs, not MDD10A. Tiers B and C require confirmed bidirectional 3S operation, independent PWM inputs, waterproof cabling, current limits and CW/CCW propeller arrangements. Voltage × current is *electrical input*; no horsepower figure can be honestly assigned without torque/rpm or shaft-dynamometer data.

5 Active GNSS, RC, telemetry and battery comparison

Table 2: Separate functions and technical evidence. Published values are attributed row by row.

Item	Electrical/physical	Interfaces, benefits and caveats	RM ea	Sourcing/evidence
Beitian BN-880 *active GNSS patch* + compass [5], [6], [7]	28× 28× 10 mm, approx 10 g; 3–5.5V family, 50mA capture typical (secondary docs)	Integrated active GNSS patch and I^2C compass claimed; serial GNSS output, compass on separate lines; vendor/board revision identification required	118.42 voucher	Shopee; maker; datasheet mirror
GI separate *active* GPS antenna [15]	26 dB gain; 50 Ω; 18 mA typical; 3m cable	SMA/IPX cable supplied; **requires compatible active-antenna RF receiver + bias**; *do not* connect to BN-880 by assumption	27.00	GI Electronic; receiver extra, not costed
M10 + IST8310 GNSS/compass [30]	exact antenna active/passive **NR**; mass/size NR	Alternate GNSS, but does *not* satisfy active-antenna requirement until antenna/LNA confirmed; cable pinout must be checked	125.09 voucher	Shopee
HotRC 6ch 2.4 GHz RC Tx + Rx [31]	Tx 230 g; receiver 4–9V; receiver mass NR	Six RC channels, fail-safe listed; packet protocol/receiver output interface confirm; as checked on 5 Nov backorder	109.00	Cytron, transmitter and receiver included
FlySky FS-i6X + receiver [32]	2.4GHz; size/mass exact chosen bundle NR	Alternative if HotRC unavailable; receiver *variant* and compatible iBUS/SBUS to be checked	207.98 display range example	Shopee; displayed entry may not include receiver
3DR/SiK **RF telemetry pair** + antennas [12], [14]	433/915MHz; listed 100mW–1000mW variants (*not* all selectable at minimum price)	MAVLink over UART to laptop ground radio; compatible TX/RX pair, frequencies and antennas must match; verify local RF rules	**160.00 A**	Shopee: 92.40–462 range, pair price NR; ArduPilot
Cytron 3S 2.2 Ah LiPo [33]	11.1V nominal; 30C advertised; 24.42Wh *derived*	Testing pack; at 80% usable, 120-min energy budget ≤9.77 W average *(calculation)*, not a runtime claim	75.00	Cytron
Cytron 3S 5.2 Ah LiPo [34]	11.1V nominal; 30C advertised; 145× 45× 28 mm; 57.72Wh *derived*	At 80% usable, 120-min energy budget ≤23.09 W average *(calculation)*; motor draws unknown	155.00	Cytron
Shopee XN Eagle 4S 14.8V 4200mAh LiPo [35]	14.8V nominal, 16.8V fully charged, 55C marketed; dimension and mass for 4S variant NR	62.16Wh nominal derived; at 80% usable, 120-min overall average ≤ 24.86 W (arithmetic only); 4S variant price NOT isolated from 2S–6S dropdown	174.67 *A*	Shopee; 4S exact SKU and inventory to confirm
Cytron B6AC 1–6S balancing charger [36]	100–240VAC or 11–18VDC; max 5A charge; 50W charge power; 575 g; 133 by 145 by 35 mm	Balance charges 4S pack on shore; configurable LiPo chemistry, terminal/lead type and safe current must be checked	95.00	Cytron; maker/distributor specs

Item	Envelope/power	Evidence	RM	Source
Cytron 2–3S balance charger [37]	110–240V AC, shore-only	Balanced charging, not boat payload; charger model and charging rate must match pack	19.90	Cytron

RF legal/integration gate: choose the 433 or 915MHz radio variant and configured transmit power only after checking MCMC/SIRIM conditions for intended field use; *no permission or licence exemption is asserted here.* The boat *may* receive waypoints or commands over MAVLink; ACK/HEARTBEAT, link drop-outs and failsafe behaviour need bench and shore trials [12], [13], [16]. **Manual override** must remain independent of MAVLink link availability.

6 Selection rationale: evidence, interfaces and design priorities

Selection principles. Minimise electronics mass, reserve one physical port for the payload MCU, keep an independent safety link, use an active GNSS receive front-end, match motor chemistry to battery voltage, and prioritise testable specifications over a low advertised Shopee variant price. The controller must support Rover differential thrust and live MAVLink communications, not merely "a GPS module and two motors" [3], [4], [10], [16]. The 20 m by 20 m, 120-minute mission is a requirement from the project brief, not a performance certificate [1].

Design decision	Technical justification and cited evidence	Condition for acceptance
CoreWing Mini V2 rather than 60A quad stack	37×26×14 mm, 19 g; 2–6S input, GPS UART5, I^2C compass and Wi-Fi MAVLink; 5V 2A continuous peripheral rail (manufacturer board integration via ArduPilot) [10].	Confirm Rover firmware, accessible connector pinout, free serial interfaces and correct ESC PWM output.
3S for short dry tests; 4S only with qualified motor	Cytron 3S packs are 11.1 V nominal / 12.6 V charged. Proposed M060 seller family specifies 12–24 V; 4S nominal 14.8 V and charged 16.8 V is within that advertised family range but exact motor variant is still not validated [24], [33], [34], [35].	Test actual battery voltage span, motor current and ESC limit before connection.
Two independent radio links	Manual RC permits local takeover; SiK-style telemetry carries two-way MAVLink for mission updates and health data. ArduPilot documents distinct receiver and GCS failsafes [9], [12], [38].	Prove RC receiver protocol compatibility and separate link-loss actions.
Active GNSS + compass	BN-880 supplier variant combines GNSS antenna, UART position and compass interface; CoreWing provides UART5 + I^2C but not onboard GNSS [5], [10], [39].	Verify active patch physically, wiring, outdoor GPS fix and magnetic installation.
Two pontoon tubes	Nominal 110 mm diameter and 1 m length makes a modular light hull candidate; displacement is calculated, not seller-rated cargo capacity [40], [41].	Full-system float, freeboard, static stability and leak tests.

6.1 Flight controller: why this board and not the more expensive stack?

The CoreWing F405 Wing Mini V2 uses an STM32F405RGT6 at 168 MHz and integrates a voltage/current monitor, 12 PWM motor/servo outputs, a GPS connector (UART5 plus I^2C), telemetry, RC input and an ESP8685 wireless interface. ArduPilot specifies 2–6S input (7–28 V), a 5 V / 2 A continuous peripheral BEC and a 5/6 V servo BEC rated 4 A continuous for the Mini V2 [10]. These are *board ratings*, not overall ship-load ratings. Its 19 g mass and 37×26×14 mm size reduce the electronics packaging penalty compared with a larger board [10]. The cited Shopee price remains variant-dependent [17].

For reference, the SpeedyBee F405 V4 60A stack has a manufacturer-documented FC plus four-channel quad ESC, but neither a 60 A ESC rating nor support for ArduPilot entails reversible propulsion. A differential-drive boat needs two motors that can physically reverse and whose ESCs interpret the Rover control signal correctly. The included quad ESC was therefore **not** used as a verified replacement for two reversible marine ESCs [4], [19], [20].

Firmware gate: flash the board-specific Rover target and confirm firmware memory-feature limitations before purchasing optional sensors [10].

6.2 Proposed port and power allocation (pinout to be checked on actual board)

Device	Suggested FC connection	Why / gate
Active BN-880 GNSS	GPS1 on UART5; compass SDA/SCL on I^2C	CoreWing manufacturer integration explicitly assigns GPS UART5 and shared I^2C. Verify TX/RX crossing, pin order and 4V5 supply [10], [39].
Independent RC receiver	Dedicated SBUS/PPM or supported serial RC on UART6	**Not the individual PWM pins of the HotRC receiver unless a supported PPM encoder is included.** Verify actual protocol, failsafe and receiver supply voltage [8], [10], [11].
SiK UART telemetry	Candidate SERIAL4 / UART4 (after pin access check)	Configure a MAVLink serial protocol and supported baud rate. Board labels and physical harness must be confirmed [10], [13].
Payload MCU	Candidate SERIAL3 / UART3 (after pin access check)	Physically separate MCU, short protected serial cable, common signal reference or isolation as required, MAVLink position/time/status transfer [10], [16].
Built-in Wi-Fi	SERIAL1 tied to wireless module	Used only as alternative development telemetry; it does not remove GPS, RC or payload MCU port needs [10].
Twin reversible ESCs	Two PWM-capable servo outputs	Set Rover skid-steer mixing, neutral deadband, reverse direction and failsafe outputs; do not confuse bidirectional DShot *telemetry* with propeller reversal [4], [10], [29].

Power-rail and noise check. CoreWing warns that its servo rail already has an onboard BEC; connecting a marine ESC's BEC red wire without checking board jumpers can backfeed supplies. The board also recommends a 35 V, 470 µF capacitor on battery input to reduce switching noise affecting its current monitor [10]. Do not assume the Mini's regulated 5 V rail can supply an unknown payload computer: the payload MCU has an independent power budget. Battery current sensing must be calibrated with a meter. A power distribution design must separately size motor-circuit fuses, ESC wiring, FC supply and grounds [10], [42], [43].

7 Battery selection: voltage, energy and realistic endurance

7.1 What the supplier actually specifies

The registered Cytron listings state that both proposed 3S LiPo batteries have 11.1 V nominal voltage, 12.6 V at full charge and a marketed 30C discharge rating. The 2.2 Ah battery measures 105×33×21 mm and costs RM75; the 5.2 Ah battery measures 145×45×28 mm and costs RM155. *Neither listing provides a validated mass, internal-resistance curve or field discharge profile* [33], [34]. Cytron warns these packs have no built-in protection against short circuits and need dedicated LiPo charging [33], [34].

The Shopee XN Eagle 4S option is advertised as 14.8 V nominal, 4.2 Ah and 55C, but it is a **multi-variant listing**: exact 4S mass, dimensions, connector and final price must be obtained from the selected checkout variant. The current RM174.67 is still a **planning allowance**, not a manufacturer-verified quote [35]. A 4S LiPo reaches 16.8 V fully charged as a cell-count calculation; this is incompatible with an ESC or motor capped at 16.0 V, even if the nominal voltage seems appropriate. Therefore Tier B is barred from 4S connection until the exact T60 version confirms an absolute input rating above the fully charged pack [22], [23].

Battery / sourcing	Datasheet/listed values	Nominal energy *(D)*	Max mean W for 2 h *(D)*	Engineering role and caution
Cytron 3S 2.2 Ah [33]	11.1 V, 2.2 Ah, 30C	24.42 Wh	9.77 W	Least-cost short bench test; cannot assume 2-h propulsion, GNSS and telemetry.
Cytron 3S 5.2 Ah [34]	11.1 V, 5.2 Ah, 30C	57.72 Wh	23.09 W	Better capacity, but insufficient verified operating voltage for nominal 12–24 V underwater thrusters.
Shopee 4S 4.2 Ah [35]	14.8 V, 4.2 Ah, advertised 55C	62.16 Wh	24.86 W	Only candidate 4S pack after variant verification; T60 16 V limit remains unresolved.

The energy values are derived by $E_{\text{nom}} = V_{\text{nom}}C$ using nominal voltage and ampere-hours from cited supplier pages. The mean-power limits use the explicit *engineering assumption* that 80% of nominal energy is available to the boat throughout a two-hour mission:

$$P_{\text{mean,max}} = \frac{0.8\,V_{\text{nom}}C}{2\text{ h}}.$$

This is not a measured state-of-health correction, weather reserve, or a guarantee of two-hour operation. With a 4S 4.2 Ah pack, a measured *total* boat draw of 10, 20, 25 or 40 W would give idealised runtimes of 4.97, 2.49, 1.99 or 1.24 h respectively under this 80% assumption; actual performance needs a discharge test at the load profile [35].

7.2 Why capacity, C-rating and peak motor current are separate criteria

The advertised 30C ratings mathematically imply $30 \times 2.2 = 66$ A and $30 \times 5.2 = 156$ A; the Shopee 55C/4.2 Ah label mathematically implies 231 A. **These are marketing-derived theoretical figures, not tested allowable continuous currents** and do not replace cell temperature, connector, cable, fuse or ESC limits [33], [34], [35]. The design current must come from the *actual submerged motor load*, including turning and transient acceleration, plus controllers, radios and payload-interface electronics.

Sizing sequence. First measure cruise, turn, start and static-maximum currents for each propeller/ESC combination at the intended immersion. Then log $P(t) = V(t)I(t)$ and integrate $\int_0^{7200\,\text{s}} P(t)\,dt$ to estimate mission Wh. Add return-to-shore reserve and battery ageing margin. Choose the pack only after the measured maximum demand and mission energy both fit the pack *and* voltage range. The initial 2-hour gate should fail if only calculated nominal Wh are presented [1], [38].

Charging/safety selection. The RM19.90 Cytron 2–3S charger is suitable only for relevant 2–3-cell chemistries, so it cannot charge a 4S battery [37]. The RM95 Cytron B6AC permits 1–6 LiPo cells, 0.1–5.0 A setting and a 50 W maximum charge rate, which makes it a compatible class of *shore charger* for a verified 4S pack, subject to correct balance lead, charge chemistry and current setting [36]. Set up battery monitoring and tested low-voltage/current failsafes, including a physical isolation switch and branch fuses; ArduPilot's low-voltage/capacity failsafes require accurate battery measurements [10], [38], [42], [44].

8 Active GPS and antenna selection: reception is not navigation accuracy

8.1 BN-880: reason for selecting a combined module

The listed BN-880 combines a GNSS receiver, antenna patch and magnetic compass in one small housing, saving a separate receiver/antenna/compass assembly. The BN-880-series pinout identifies six connections: compass SDA/SCL, GNSS TX/RX, supply and ground; hence it can connect directly in *interface terms* to CoreWing UART5 plus I^2C, with the exact connector/harness confirmed before power-up [5], [10], [39]. A reproduced BN-880 specification reports approximately 28×28×10 mm, approximately 10 g in one reseller listing, a nominal 5 V supply and approximately 50 mA acquisition draw at 5 V; $5(0.05) = 0.25$ W is a *derived upper-level planning figure*, not an independent consumption measurement [39], [45]. The listing's *active-antenna* assertion must be checked against the actual module label/circuit, since Shopee variations and unofficial reproductions do not constitute traceable manufacturer testing [5], [7].

For technical provenance, Beitian's *official* BE-880 (note the extra letter "E": **not the BN-880 SKU**) is published at 28×28×11 mm, 12 g, 3.6–5.5 V and 15 mA typical at 5 V using an M10050 receiver. Those numbers *must not be transferred to BN-880* without confirming the received device is actually the BE-880 [46]. This comparison

demonstrates why the report states model and revision rather than copying attractive GNSS numbers between different Beitian products.

8.2 Antenna options and why adding "26 dB" is not automatic improvement

Approach	Traceable evidence	Suitability / risk
Integrated BN-880 patch (baseline)	Shopee GNSS/compass listing and BN-880-series pinout; antenna amplification asserted, not independently RF-tested [5], [7], [39].	Compact, few connectors, low cable loss; mount with clear sky view above splashes and away from high-current motor wiring.
GI K0500001 separate active antenna	Registered supplier specifies RM27, 26 dB advertised gain, 50 Ω output, 18 mA typical current, 3 m cable and IPX–SMA adapter [15].	Requires a *different compatible GNSS receiver* with active-antenna bias and matching RF connector; not electrically connectable to a sealed BN-880 by assumption.
Shopee M10 + IST8310 compass	Seller lists M10 receiver with external compass but does not substantiate active antenna/LNA, receiver/antenna gain, cable or exact mass for a chosen variant [30].	Only qualify if maker documents active RF front-end and actual received board; not an automatic substitute for mandatory active GNSS.

A "26 dB" advertised antenna gain does *not* establish final GPS accuracy or range: the receiver must supply suitable antenna bias, impedance must match, and a 3 m feed adds frequency-dependent loss. The GI listing provides no matched-receiver validation, waterproof connector rating, or *system-level* accuracy test [15]. The separate antenna is therefore an option for a properly designed receiver front-end, not a budget line that can simply be added to the BN-880.

8.3 GNSS performance requirement for a 20 by 20 m grid

The BN-880 secondary specifications describe approximately 2 m horizontal position accuracy under favourable open-sky assumptions [45]; Beitian's separate official BE-880 lists 2.0 m CEP, but again this is *different hardware* [46]. The project uses 5 m nominal lane separation as an engineering planning value. On a compact surface boat, multipath from banks, compass errors, crosswind and finite control response may consume the spacing margin, so **5 m repeatable lane accuracy has not been demonstrated** by the module's nominal figure alone. Compass readings must be calibrated with thrusters on and off; motor and ESC interference is a recognised reason that ArduPilot supports dual-GNSS yaw alternatives [47].

Acceptance test: log GNSS fix quality, horizontal dilution/accuracy estimator, compass heading, ground-track cross-track error, and repeated waypoint returns for multiple directions. Keep time-stamped raw logs and compare against marked shore reference points; if the resulting cross-track distribution erases coverage, change spacing or upgrade positioning before claiming full spatial coverage. The boat FC passes timestamped position and fix metadata to the separate payload MCU; GNSS hardware is *not* part of the payload budget [1], [10], [16].

9 Radio modules and antennas: three independent jobs

9.1 Manual RC: why it remains mandatory even with autonomy

The Cytron HotRC kit includes a transmitter and receiver and lists 2.4 GHz FHSS, six channels, 4–9 V receiver power, PWM response at or below 20 ms, approximately 230 g *transmitter mass*, and advertised 300 m ground/800 m air distances [8]. The distances are supplier marketing values, **not a measured over-water range**. A six-channel link is useful for steering, throttle, mode selection and an independent stop; the exact channel assignments are configuration choices [9], [38].

Critical compatibility correction to V3.1. Cytron does *not* establish that the HotRC receiver exports a combined SBUS, PPM or compatible serial protocol. A 6-channel *individual PWM* receiver does not plug into CoreWing's single SBUS/PPM input as if it were one combined control stream. ArduPilot explicitly describes PPM encoders for old-style parallel PWM receivers [8], [10], [11]. **Purchase gate:** request the receiver model/pinout and confirm a single combined signal, or add a registered-supplier/Shopee-sourced encoder and its price, or select a transmitter + receiver bundle with documented compatible serial output. Until then the RC part and hence every displayed tier subtotal is conditional. The Shopee FlySky alternative has transmitter/receiver variants; a price for *transmitter only* must not be used as the price of an operable RC link [32], [48].

9.2 RF telemetry: why SiK rather than using the RC channels for everything?

For ground-station status, waypoints and parameter changes, use a transparent two-way serial channel carrying MAVLink. ArduPilot's SiK reference describes small radio modems in 433 and 915 MHz variants, under 4 g *without antenna*, up to 20 dBm / 100 mW in the documented reference platform, sensitivity down to -121 dBm and configurable serial/air data rates (up to 250 kbit/s) [12]. These are *reference-platform* specifications, **not** verified performance or frequency approval for the selected Shopee 100/500/1000 mW multi-variant listing [14]. The boat unit needs a UART, regulated compatible power and antenna; the ground unit needs a matched antenna and USB interface to Mission Planner [12], [13].

In a 20 by 20 m test area, onboard CoreWing Wi-Fi can carry TCP/UDP MAVLink without additional RF hardware [10]; this saves the provisional RM160 SiK-pair allowance. However, no board-specific Wi-Fi antenna gain, water-level propagation plot, outdoor packet-loss rate or link range is published in the cited board documentation. Wi-Fi is a *development alternative*, not proof of a robust survey control link. Retain RC manual override regardless [10], [38].

Antenna bill of materials: one antenna on the RC *receiver* and its matching transmitter aerial (normally bundled), plus one antenna per SiK radio at the boat and shore. The exact Shopee radio purchase must explicitly include *two modems, two correct-band antennas, USB/UART cable and connectors* or the cost table must add missing items. Do not interchange 433 and 915 MHz antennas, infer gain or range from their appearance, or assume the seller's high-power option shares the ArduPilot 100 mW reference characteristics [12], [14]. The correct Malaysian frequency band and transmit power require a documented MCMC/SIRIM compliance check before deployment; this report does not claim approval.

Placement rationale: position GNSS patch facing the sky; stand RC/RF aerials clear of water, battery and conductive motor wiring; route power cables together and separate them from the compass lead where physically practical. These are noise-reduction design decisions that need a field link-budget and compass-disturbance test; no numerical radio-range claim follows from these recommendations [10], [12], [47].

9.3 Loss-of-link behaviour: a safety case, not a checkbox

ArduPilot distinguishes RC receiver loss from ground-station MAVLink heartbeat loss. The Rover documentation gives a default RC timeout of 1 s and default GCS heartbeat timeout of 5 s; neither default should be taken as a verified safe action. A lost RC link may result in Hold, RTL or Disarm according to parameters, while GCS loss may cause a different behaviour [38]. **Dry-test each combination** of RC loss, telemetry loss, GPS unavailable and low battery; only allow autonomous motion when the independently testable chosen response is safe for the environment. Hardware kill-switch operation remains separate from these software failsafes [38], [44]. **Do not advertise RC as a guaranteed emergency stop** until observed receiver failsafe output and ArduPilot modes have been tested.

10 Propulsion, waterproofing and small components: justification ledger

Table 3: Why each remaining boat-common component is included; product figures are source-linked, and selection gaps remain explicit.

Component	Engineering reason for including it	Required evidence / unresolved qualification
Two underwater motors and propellers [21], [22], [24]	Differential thrust makes turning possible without a separate rudder servo, and provides independent port/starboard actuation with Rover motor mixing [3], [4].	Verify motor type, both directions, CW/CCW propeller geometry, thrust at cruise, wet current, seals.
Cytron MDD10A (Tier A only) [26]	Dual brushed motor driver; 5–30 V motor input, 10 A continuous and 30 A / 10 s per channel; 84.5×62 mm. Useful only if the two-wire motors are brushed.	Motor type and peak current; **PWM/DIR is not RC-servo PWM**. Include verified interface translation.
T60 kit ESC (Tier B) [22], [23]	Intended integrated BLDC drive concept, but listing variants may be bare motor, ESC or bundle. "300 W" is not proof of shaft power or 120-minute cruise load.	Exactly sold variant, reverse firmware, 4S fully charged voltage, heat/current test.
ZTW Shark G2 20 A ESC (Tier C) [28], [29]	OEM specifies 20 A continuous / 30 A burst, 2–4S, 60×33×10 mm, 32 g and adjustable 5/6 V, 4 A BEC. Reversible marine BLDC alternative.	Verify 20 A exact seller SKU, wet M060 current < 20 A continuous, separate ESC BEC red lead, thermal response.
CoreWing board built-in regulator [10]	Supplies lightweight boat navigation electronics, reducing external conversion boards in early prototypes.	Track 5 V peripheral current vs 2 A board rating; prevent servo BEC backfeeding.
LiPo charger [36], [37]	Enables supervised cell-balanced shore charging; 4S build needs a 1–6S unit, not a 2–3S charger.	Match battery connector, balance port, chemistry and configured current.
UPVC pontoons and end caps [40], [41]	Modular float tubes with a nominal cylindrical displacement; twin geometry offers initial transverse stability and a platform between floats.	Verify actual pipe OD, cap fit, total mass, freeboard, yaw stability and leaks.
PVC solvent cement [49]	Seals compatible UPVC cap joints; cheap fastening method with little additional hardware.	Check chemical compatibility, full cure, leak-test without payload.
IP65 electronics enclosure [50]	Splash protection and orderly mounting of low-voltage electronics. IP65 is *not* an immersion rating.	Protected deck placement, gasket inspection, cable-gland entry and splash test.
Cable glands [51]	Mechanical sealing and strain relief where cables enter the electronics enclosure.	Choose gland by actual cable diameter, check strain relief and leak path.
Fuses and inline holders [42]	Protect wiring from damaging fault currents; separate branch fuses prevent one short from bypassing protection.	Choose fuse according to *wire ampacity and measured draw*, not seller's maximum advertised fuse.
Wire and connectors [43]	Electrical distribution between battery, two drivers and avionics; cable cross-section affects loss and overheating.	Do not assume AWG18 is acceptable for motor branches; calculate voltage drop and validate sustained current and connector rating.
Manual isolation switch [44]	Physical battery disconnection during recovery/bench work, independent of software state.	Confirm DC contact rating, wiring, accessible mount and wet-hand operation.

11 Acceptance and procurement gates by component

Table 4: Required evidence before paying for a full build or claiming a 120-minute mission.

Subsystem	Minimum test or source document	Current status
RC receiver protocol	Seller/manufacturer confirms SBUS/PPM/serial RC on exact bundle; otherwise itemised approved-source PPM encoder, then confirm RC calibration [8], [10], [11].	**Open: unqualified**.
GNSS	Exact variant's active antenna/LNA evidence, 5 V compatibility, UART/I^2C pinout, measured GNSS and compass performance [5], [7], [10].	Open.
Radio pair/antennas	Matched legal-band two-unit SiK kit, boat and ground antennas, cable/connector pinouts and laptop MAVLink command round trip [12], [14].	Open; RM160 allowance.
Battery	Exact pack SKU, weight, connector, charge voltage, thermal and voltage sag, sustained power consumption vs 120-minute objective [33], [34], [35].	Open; 4S price allowance.
ESC/propulsion	Correct motor type; reverse enabled; current-vs-thrust bench graph; fully charged pack voltage inside all device ratings [22], [24], [26], [29].	Open; Tier B 16 V conflict.
FC power and UART map	Physical harness mapped to manufacturer docs; no ESC BEC rail backfeed; stable GNSS/RC/SiK/payload MCU simultaneous logging [10].	Open.
Hull, leakage and mass	Weigh assembled system, ballast test with actual payload, freeboard and righting/stability measurement, dry electronics after spray [1], [40], [50].	Open; no measured displacement.
Fail-safe	Physical isolation; deliberately lose RC, SiK/Wi-Fi, GNSS; observe configured Hold/RTL/Disarm; current + voltage monitor under load [10], [38], [44].	Open.
Budget completeness	Quote all exact variants, antenna and RC conversion accessories, wiring, mounts, shipping and charger.	**Not yet qualified**; the tier totals are partial.

12 Three conditional increasing-cost common-core configurations

Voltage engineering correction. A 3S LiPo is only 11.1 V nominal, below the advertised 12V lower bound of the T60 and M060 thruster families. For tiers B/C, replace the original 3S 5.2 Ah pack with a **4S, 14.8V nominal, 4200mAh pack**; a Shopee XN Eagle multi-variant listing spans RM66.67–174.67, so **RM174.67 is an upper-range *allowance*, not a confirmed 4S price or stocked variant** [35]. Select the Cytron B6AC 1–6S compatible charger, **RM95** [36], since a 2–3S-only charger **cannot charge a 4S pack** [37]. *CRITICAL:* a fully charged 4S LiPo reaches 16.8 V; the T60 family sometimes quotes a 16V limit, so the motor supplier **must confirm full-charge voltage compatibility before 4S is connected** [22], [23]. This remains a design gate, not a proven electrical combination. For M060 12–24V, 4S nominal and charged voltage fall within its advertised range, subject to exact sold-SKU verification [24].

Important: These configurations do not vary by sample payload mass. All include two proposed thrusters, an ArduPilot-compatible boat FC, BN-880 *active* GNSS/compass, a **provisional** RC transmitter/receiver, a bidirectional RF telemetry pair, one LiPo, charger, pontoons, enclosure, fused wiring and isolation switch. **All tier totals omit any RC PWM-to-PPM encoder or replacement receiver, mounting and missing hardware: none is yet a complete purchase-ready build.** The tier A motor/driver pairing is only a *conditional* study until brushed type and current are confirmed. Tier B pair pricing is a *variant-bounded allowance*, not a verified kit quotation. Tier C ESC prices are allowances. Cost inputs each have purchase references in the complete cost table below.

Tier	Twin motors / drive	FC, active GPS, radio	Main battery	Rough RM
A (low)	2 × generic 12V underwater jet (*brushed? NR*) + Cytron 2-ch MDD10A [21], [26]	CoreWing F405 Mini V2; BN-880 active; HotRC RC; 3DR/SiK pair (A) [5], [14], [17], [31]	3S 2200mAh [33]	**1192.37**
B (mid)	2 × T60-type BLDC + 40A reversible ESC bundles *only if verified* [22], [23]	Same common core and antenna/links [7], [14], [17], [31]	4S 4200mAh (A) [35]	**1751.68**
C (upper)	2 × M060 12–24V BLDC + 2 × ZTW 20A ESCs (A) [24], [28]	Same common core, GPS, RC and RF [5], [14], [17], [31]	4S 4200mAh (A) [35]	**1855.40**

Tier total *incl. basic hull*	Remaining vs RM2k	Share of cap	Budget judgement
A: **RM 1192.37**	RM 807.63	59.6%	Cheapest *research budget*, but NOT purchase-ready without identifying a brushed/compatible propulsion motor.
B: **RM 1751.68**	RM 248.32	87.6%	**Conditional only**: confirm T60 motor accepts 16.8V fully charged 4S, ESC variant, and physical forward/reverse.
C: **RM 1855.40**	RM 144.60	92.8%	Very limited remaining headroom; not preferred. Confirm exact motor/ESC variants, peak current and 4S charger.

4S budgeting caveat: Both higher tiers use an estimated 4S 4200mAh price at RM174.67 from a multi-variant Shopee listing. The exact 4S variant may cost less or more and may be unavailable. The old 3S/Cytron B3AC combination is not appropriate for a motor specified to require 12V minimum throughout discharge. [34], [35], [36], [37]

Additional V3.2 RC gate: The HotRC bundle is costed but its individual-PWM versus SBUS/PPM/serial output remains unresolved, so a converter or alternative RC bundle could increase every tier cost. Radio pair price and antenna contents remain provisional.

Critical total caveat: These are planning subtotals from cited supplier pages and clearly marked allowances. Antenna mounts, correct wiring adapters, watertight fixing hardware, thruster brackets, RF approvals/shipping, propeller guards and final fabrication can change costs; battery and radio variants can change prices. The **payload MCU, sensors, pumps, filter/flow cell, sample bottles and image processing hardware are excluded**. The 2h gate is **not proven**. If the RM2,000 ceiling must include payload hardware too, a fresh combined budget is necessary once Track A/B is selected.

13 Source-audited itemised cost table (per configuration)

Table 5: Common-core cost basis: prices and assumptions attributed by IEEE reference on every row. A = allowance; voucher and variation prices can differ at checkout.

Tier	Purchasable item and source	Qty	RM ea	RM ext	Procurement status	Technical/sourcing caveat
A	CoreWing F405 Wing Mini V2 [17]	1	264.43	264.43	Observed voucher listing	Price for unspecified pin variant; verify Rover firmware
A	BN-880 GPS/compass + active patch [5]	1	118.42	118.42	Observed voucher listing	Active patch assertion uses secondary BN-880 datasheet; confirm genuine chip/compass
A	HotRC 6ch Tx + Rx bundle [31]	1	109.00	109.00	Published supplier price	Backorder at final check; inspect receiver output protocol before ordering
A	3DR/SiK air+ground radio pair [14]	1	160.00	160.00	PLANNING ALLOWANCE	Visible multi-variant range RM92.40 to RM462; PAIR and exact band NOT verified
A	2-3S LiPo balance charger [37]	1	19.90	19.90	Published supplier price	Required only if not already available

Tier	Part/source	Qty	Unit	Ext	Price type	Constraint
A	IP65 box 263x182x60 mm [50]	1	35.00	35.00	Published supplier price	Water-resistant against jets, NOT watertight on immersion
A	110mm UPVC pipe 1m [40]	2	22.90	45.80	Upper listed variant	100cm each; weight, wall thickness and hull capacity unverified
A	110mm UPVC end cap [41]	4	5.88	23.52	Observed seller listing	Fit with pipe must be checked
A	PVC solvent cement [49]	1	15.30	15.30	Upper listed variant	Set-up and watertightness tests required
A	10-piece gland kit [51]	1	7.08	7.08	Upper listed variant	Sizing by actual wire OD
A	Fuse holder and fuse [42]	3	6.85	20.55	Upper listed variant	Fuse value MUST be set by wire/current ratings
A	AWG18 by metre [43]	4	2.50	10.00	Prior listing, re-quote	Not necessarily suitable for high-current motor branches
A	12-48V battery isolator [44]	1	29.99	29.99	Observed seller listing	Seller 275A claim unverified; confirm contact ratings
A	12V 3-blade underwater jet thruster [21]	2	74.80	149.60	Upper visible variant / NOT MOTOR-SPECIFIC	Brushed/BLDC type, stall draw, CW and CCW pair unconfirmed
A	Cytron MDD10A 2ch brushed H-bridge [26]	1	108.78	108.78	Published supplier price	ONLY if motor is confirmed brushed and each channel <= 10A continuous
A	3S 11.1V 2200mAh LiPo [33]	1	75.00	75.00	Published supplier price	Short-duration bench/RC testing only; 2h not established
B	CoreWing F405 Wing Mini V2 [17]	1	264.43	264.43	Observed voucher listing	Price for unspecified pin variant; verify Rover firmware
B	BN-880 GPS/compass + active patch [5]	1	118.42	118.42	Observed voucher listing	Active patch assertion uses secondary BN-880 datasheet; confirm genuine chip/compass
B	HotRC 6ch Tx + Rx bundle [31]	1	109.00	109.00	Published supplier price	Backorder at final check; inspect receiver output protocol before ordering
B	3DR/SiK air+ground radio pair [14]	1	160.00	160.00	PLANNING ALLOWANCE	Visible multi-variant range RM92.40 to RM462; PAIR and exact band NOT verified
B	Cytron 1–6S multifunction balance charger [36]	1	95.00	95.00	Published supplier price	4S-capable shore charger, matching connector required
B	IP65 box 263x182x60 mm [50]	1	35.00	35.00	Published supplier price	Water-resistant against jets, NOT watertight on immersion
B	110mm UPVC pipe 1m [40]	2	22.90	45.80	Upper listed variant	100cm each; weight, wall thickness and hull capacity unverified
B	110mm UPVC end cap [41]	4	5.88	23.52	Observed seller listing	Fit with pipe must be checked
B	PVC solvent cement [49]	1	15.30	15.30	Upper listed variant	Set-up and watertightness tests required
B	10-piece gland kit [51]	1	7.08	7.08	Upper listed variant	Sizing by actual wire OD
B	Fuse holder and fuse [42]	3	6.85	20.55	Upper listed variant	Fuse value MUST be set by wire/current ratings
B	AWG18 by metre [43]	4	2.50	10.00	Prior listing, re-quote	Not necessarily suitable for high-current motor branches
B	12-48V battery isolator [44]	1	29.99	29.99	Observed seller listing	Seller 275A claim unverified; confirm contact ratings
B	Hobbyocean T60-type 16V BLDC + 40A ESC bundle [22]	2	321.46	642.92	Upper visible variant / BUNDLE UNCERTAIN	Seller variants may be motor-only/ESC-only; forward/reverse must be confirmed
B	4S 14.8V 4200mAh LiPo [35]	1	174.67	174.67	Upper listed variant / ALLOWANCE	Exact 4S price and inventory NR; 4S charged is 16.8V; verify thruster/ESC limits
C	CoreWing F405 Wing Mini V2 [17]	1	264.43	264.43	Observed voucher listing	Price for unspecified pin variant; verify Rover firmware
C	BN-880 GPS/compass + active patch [5]	1	118.42	118.42	Observed voucher listing	Active patch assertion uses secondary BN-880 datasheet; confirm genuine chip/compass
C	HotRC 6ch Tx + Rx bundle [31]	1	109.00	109.00	Published supplier price	Backorder at final check; inspect receiver output protocol before ordering
C	3DR/SiK air+ground radio pair [14]	1	160.00	160.00	PLANNING ALLOWANCE	Visible multi-variant range RM92.40 to RM462; PAIR and exact band NOT verified
C	Cytron 1–6S multifunction balance charger [36]	1	95.00	95.00	Published supplier price	4S-capable shore charger, matching connector required
C	IP65 box 263x182x60 mm [50]	1	35.00	35.00	Published supplier price	Water-resistant against jets, NOT watertight on immersion

Tier	Part/source	Qty	Unit	Ext	Price type	Constraint
C	110mm UPVC pipe 1m [40]	2	22.90	45.80	Upper listed variant	100cm each; weight, wall thickness and hull capacity unverified
C	110mm UPVC end cap [41]	4	5.88	23.52	Observed seller listing	Fit with pipe must be checked
C	PVC solvent cement [49]	1	15.30	15.30	Upper listed variant	Set-up and watertightness tests required
C	10-piece gland kit [51]	1	7.08	7.08	Upper listed variant	Sizing by actual wire OD
C	Fuse holder and fuse [42]	3	6.85	20.55	Upper listed variant	Fuse value MUST be set by wire/current ratings
C	AWG18 by metre [43]	4	2.50	10.00	Prior listing, re-quote	Not necessarily suitable for high-current motor branches
C	12-48V battery isolator [44]	1	29.99	29.99	Observed seller listing	Seller 275A claim unverified; confirm contact ratings
C	M060 underwater BLDC thruster 12-24V [24]	2	273.32	546.64	Observed voucher listing, 2 variants	No validated thruster input draw or thrust for exact variant
C	ZTW Shark G2 20A bidirectional ESC [28]	2	100.00	200.00	PLANNING ALLOWANCE	Listing RM67.65-227.70 over all variants; 20A exact not quoted; motor <=20A unproven
C	4S 14.8V 4200mAh LiPo [35]	1	174.67	174.67	Upper listed variant / ALLOWANCE	Exact 4S price and inventory NR; 4S charged is 16.8V; verify thruster/ESC limits

14 Mass, energy, buoyancy and validation gates

The hull concept reuses two 110 mm nominal-diameter 1 m UPVC pontoons, end-capped and solvent-bonded [40], [41], [49]. **Ideal external cylindrical displacement** (not verified load capacity) is $V_{\text{ext}} = 2\pi(0.055\,\text{m})^2(1.0\,\text{m}) = 19.0\,\text{L}$, calculated using the listed nominal diameter and nominal length; real waterplane, wall thickness, caps, freeboard and stability *reduce the usable margin*. At half immersion the simplistic submerged volume is approximately 9.5 L, so it is unsafe to infer 9.5 kg useful payload: *the entire boat mass* consumes this displacement. The V3 Method Rationale's stored-water scenarios cannot be assigned to this hull without a redesigned float geometry and direct flotation testing.

For 120-min operation, the new Tier B/C 4.2 Ah 4S pack has $14.8\,\text{V} \times 4.2\,\text{Ah} = 62.16\,\text{Wh}$ nominal (derived from the advertised cell count and capacity) [35]. If the *engineering assumption* is that only 80% of nominal energy is usable, the entire boat load must average $\leq 62.16 \times 0.8/2 \approx 24.86\,\text{W}$ to operate for 2 h, *without weather/current margin*. The Tier A 2.2 Ah 3S pack similarly provides only $11.1 \times 2.2 = 24.42\,\text{Wh}$ nominal, or $24.42 \times 0.8/2 = 9.77\,\text{W}$ over two hours [33]. These are **battery-only arithmetic constraints**, not measured endurance. Motor load curves and bridge efficiency remain unknown [21], [22], [24]. Two independent 3S packs are *not* assumed to be safely paralleled.

Pass/fail gates before any full-water trial:

1. **Procurement:** prove correct purchased motor CW/CCW variants, brush type, driver mode, ESC kit inclusion and exact checked-out price; verify RC receiver and air/ground radio pair include antennas.
2. **Dry electronics:** test 3S powered flight controller, active GNSS fix, compass away from thruster leads, independent RC disarm and MAVLink ground-station mode/waypoint command.
3. **Power protection:** check wire and fuse ratings; test battery isolator, BEC topology, common grounds, telemetry pins and potential ESC BEC backfeed.
4. **Waterproofing:** static leak test, float/freeboard and full-mass stability tests; test separately with each expected Track A/B payload installed, then assess turning and recovery in tethered water trials.
5. **Endurance:** log current at idle, straight cruise, turning and maximum thrust; measure usable battery Wh, then run 120 min with margin under representative conditions.
6. **Georeferencing:** confirm Boat FC sends timestamp, GPS fix status, latitude/longitude and movement state to the independent payload MCU over a dedicated serial MAVLink feed. Check actual lane-spacing errors in water.

15 Revised procurement order and final recommendation

Do not first purchase a SpeedyBee 60A stack or 4S battery before verifying motor voltage. Start by confirming the CoreWing controller Rover firmware and pinout, the exact BN-880 module with active GNSS antenna and compass, an **ArduPilot-compatible** manual 2.4GHz RC receiver output, and a legal-band SiK-style

RF pair; do not assume the listed HotRC PWM receiver is directly compatible [7], [10], [12], [31]. This provides an isolated navigation and communication backbone independent of whether V3 later selects Track A or Track B.

No tier is procurement-ready as written. Tier A is the lowest-cost *bench concept* but requires a confirmed brushed motor, motor-current test, and RC protocol solution. Tier B is potentially the lowest-cost capable BLDC package, but remains conditional on T60 kit contents, full 4S-charge voltage tolerance, reversible operation and submerged current. Tier C has a clearer advertised 12–24V input range at 4S, but it consumes almost all headroom and its motor current is unverified. Select a tier *only after* each gate in Table 4 is passed; none presently proves two-hour endurance. **No tier has validated 2h runtime or 1kg+ useful payload capacity.**

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